The Cyber Risk Premium

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First Draft: June 28, 2020
This Version: July 4, 2022

Abstract

Cyber risk is an important but latent source of risk in the economy. To estimate its impact on the asset market, we use machine learning techniques to develop a firm-level measure of cyber risk. The measure aggregates information from a rich set of firm characteristics, and shows superior ability to forecast future cyberattacks on individual firms. We find that firms with higher cyber risk earn higher average stock returns, which cannot be explained by standard asset pricing models. When these firms underperform, cybersecurity experts tend to have higher concerns about cyber risk and the cybersecurity exchange-traded funds outperform.

JEL classification: G11, G12, G14, G17.

Keywords: Cyber risk, Cybersecurity, Risk premium, Machine learning, Textual analysis.

We thank Andrea Eisfeldt, Nuri Ersahin, Wayne Ferson, Itay Goldstein, Larry Harris, Gerard Hoberg, Ryan Israelsen, Mete Kilic, Dmitriy Muravyev, Markus Pelger (discussant), Vincenzo Quadrini, David Robinson, Selale Tuzel, Michael Weber, Ivo Welch, and participants in seminars held at the Michigan State University and University of Southern California, and the Winter Research Conference on Machine Learning and Business for valuable comments and suggestions.

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1. Introduction

The digital transformation of the economy and increased interconnectivity have created unprecedented opportunities and under-explored risks. On the one hand, the ever-growing pool of data along with advances in artificial intelligence helps to promote efficiency and productivity in the economy; on the other hand, it exposes households, businesses, and governments to a new and potentially systemic source of risks: the cyber risk.[1] In the past decade, various forms of cyberattacks have attracted widespread attention. For instance, the Equifax data breach in 2017 exposed approximately 147 million names and dates of birth, 145.5 million Social Security numbers and 209,000 payment card numbers along with expiration dates, which led to a settlement of nearly $700 million with federal and state investigators. The recent SolarWinds cyberattack, which came to light in December 2020, illustrates both the vulnerability of the most secure networks to such attacks, and the potential for immense collateral damage to corporate and governmental entities connected to these networks.

Considering the importance of cyber risk, it is natural to study how it might influence stock market prices and returns. The primary challenge for such a study is that cyber risk is latent, not directly observable. To address this challenge, we use machine learning algorithms to develop a real-time estimate of the likelihood for each individual firm to experience a cyberattack in the subsequent period. The underlying logic is that hackers do not choose their targets at random, but focus on firms with certain attributes (see, e.g., Kamiya et al. (2021)). Moreover, firms tend to communicate their self-perceived exposures to cyber risk through their disclosures, e.g., through the 10-K filings. This information is also likely to be important in predicting the level of the firms' cyber risk. Machine learning techniques are particularly suited to this task due to their ability to extract useful information from a large set of features and to deliver superior out-of-sample forecasts.

In our empirical analyses, we introduce a novel data set of cyberattacks, and apply a variety of machine learning techniques including the logistic ridge regression, and the K-Nearest Neighbor and Naive Bayes combined with an "EasyEnsemble" sampling technique (EEC-KNN and EEC-NB).[2] We find that our cyber risk measure based on these algorithms has superior ability to forecast the occurrence of future cyberattacks. For instance, compared with the simple logistic forecasting model, the predictive performance of the logistic ridge

[1] See Kashyap and Wetherilt (2019) for discussions on the distinguishing features of cyber risk.

[2] The EasyEnsemble technique combines undersampling and bootstrapping, which is effective in dealing with class-imbalance problems. This technique is particularly suited to address issues like cyberattacks, which represent rare events in the data. See Section 5.2 for relevant technical details on the EasyEnsemble technique.

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regression improves from 0.4% to 6.3% based on the harmonic mean of precision and recall rates (F-score) and from around 1.4% to 59.9% based on the geometric mean of sensitivity and specificity (G-mean). Because the different machine learning techniques yield similar results, we present our main results using the logistic ridge regression, which is commonly used in the asset pricing literature.

Armed with the cyber risk measure, we study how cyber risk is related to stock returns. Our primary tests use the Fama and MacBeth (1973) cross-sectional regressions to examine the incremental predictive power of the cyber risk measure for stock returns against well-known firm characteristics in the period from July 2008 to June 2019. In the first set of regressions, we include the natural log of market capitalization, book-to-market ratio, gross profitability, investment ratio, and past 11-month return skipping the most recent month as control variables. In the second set, we further include the past one-month return, idiosyncratic volatility (Ang et al., 2006), and the Amihud illiquidity ratio (Amihud, 2002). Finally, we also control for the organizational capital (Eisfeldt and Papanikolaou, 2013), CAPM market beta, tail risk beta (Kelly and Jiang, 2014), coskewness (Harvey and Siddique, 2000), and net operating assets (Hirshleifer et al., 2004). In all these regressions, the relation between cyber risk and stock returns is economically meaningful and statistically significant. In terms of magnitudes, a one standard deviation increase in cyber risk is associated with higher average stocks returns of between 0.16% and 0.21% per month. These results show that cyber risk is an independent and important driver of cross-sectional variation in stock returns during our sample period.

We also perform a standard portfolio analysis. At the end of each June from 2008 to 2018, we sort stocks into five quintiles based on the estimated level of cyber risk using information available in real time, and form value-weighted portfolios that are rebalanced at the end of the following June. Consistent with the Fama-MacBeth regressions, we find that stocks in the top quintile with the highest level of cyber risk tend to have higher returns than those in the bottom quintile with the lowest cyber risk. The spread in returns is 0.84% per month after we adjust for their market exposure and 0.58% per month after adjusting for Fama-French five factors and momentum. The alpha remains large and statistically significant when we form tercile and quartile portfolios as well as under alternative asset pricing models.

In addition to the cross-sectional variation in average returns, we examine the time variation in the relative returns between stocks with high and low cyber risk. We conduct two tests. First, we study the relation between the variation in the return difference between high and low cyber risk stocks and the New York University's Index of Cyber Security (ICS), which is based on monthly surveys of industry executives about their perceived level of cyber risk. Because the ICS is a monthly aggregate measure of cyber security, we form

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a factor-mimicking portfolio and compute its performance. We estimate each stock's return sensitivity to changes in ICS in the past year. A lower ICS beta is associated with higher firm-level cyber risk. Then we form a spread portfolio that buys stocks with high ICS beta and sells those with low ICS beta. The return on the spread portfolio based on our cyber risk measure and that based on the ICS beta have a strong negative comovement, with a time-series correlation coefficient of -32%.

Second, we study the relation between the performance of the spread portfolio based on our cyber risk measure and that of two cybersecurity ETFs that invest in firms providing cybersecurity services. We again find strong negative comovements between the two, with time-series correlation coefficients below -40%. That is, when stocks with higher cyber risk underperform, cybersecurity firms tend to have higher returns. These results suggest that the time variation in the return spread between high and low cyber risk firms is likely driven by the market's perception of cyber risk in the economy.

We evaluate the robustness of our results by performing additional tests. First, we split our sample into two subperiods. We find that in both periods, the cyber risk measure is positively related to future stock returns, with the effect slightly stronger in the more recent subperiod. Second, we compute industry-adjusted returns for each stock in our sample and find that the positive relation between cyber risk and future stock returns remains strong. Third, we consider different machine learning algorithms and use different dictionaries in the linguistic analyses to estimate the cyber risk measure. The results are robust. Finally, we consider a placebo test to pinpoint the importance of cyber risk. In particular, we randomly select firms to construct a fake sample of cyber attacked firms. The number of random draws to create the fake sample equals the actual number of cyberattacks for each industry in each year. We then use the same machine learning algorithms to estimate each firm's pseudo-cyber risk, and estimate its relation to stock returns. Our results show that this relation is essentially flat. This result highlights the importance of cyber risk in driving stock returns.

Our study contributes to the fast growing literature that studies the implications of cyber risk for firms, the financial markets and the economy. Many studies in this literature focus on the impact of realized cyberattacks. For instance, Kamiya et al. (2021) find that the announcement of successful cyberattacks is on average associated with a 1.09% wealth loss for shareholders within a three-day window around the incidents. They argue that successful cyberattacks lead to value loss for victims for both actual and reputational reasons.[3] No-

[3]In another study, Michel et al. (2020) study the timing of cyberattack announcements. They find negative abnormal returns on cyberattack victims prior to the announcement of the attacks, which is consistent with some information leakage. Echoing this message, Lin et al. (2020) report evidence of insider trading ahead of public announcements of cyberattacks. Binfarè (2019) studies the effect of data breaches on the cost of debt financing. He finds that lenders tend to charge borrowing firms larger spreads after they experience

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table exceptions include Jamilov et al. (2020) and Florackis et al. (2022), who use textual information to identify firms with high cybersecurity risk. Jamilov et al. (2020) use firms' earnings conference call transcripts to develop a measure for firm-level cyber risk exposure and sentiment, and provide evidence that the risk is potentially systemic.[4] However, they do not focus on the pricing of cyber risk in the stock market. Closer to us, Florackis et al. (2022) study the relation between cyber risk and stock returns. They use firms' descriptions of cyber-related risk in 10-K reports to identify firms with high cyber risk. Their conclusion that cybersecurity risk is priced in the cross-section of stock returns provides independent support for our results. The main difference between the two studies is methodological. Florackis et al. (2022) compare the word distribution of "Item 1A. Risk Factors" in the 10-K reports of training firms with that of the hacked sample, identifying the most similar firms as high cybersecurity risk firms. Their method is simple and powerful, relying only on the text data in firms' annual reports. Our machine learning algorithms provide a mapping from any feature set to the occurrence of cyberattacks. The special strength of our approach is its flexibility: the feature set can be expanded to encompass any forms of data, including traditional and alternative data.[5]

Our paper also joins the burgeoning literature that applies machine learning techniques to asset pricing. This literature has focused on using machine learning algorithms to efficiently select and combine firm characteristics to predict stock returns (see, e.g., Freyberger et al. (2020); Gu et al. (2020)) and to construct a robust stochastic discount factor (see, e.g., Kelly et al. (2019), Kozak et al. (2020), and Chen et al. (2020)). Instead of directly targeting asset returns, our paper uses machine learning techniques to estimate an important yet latent risk as economies become more digital. Our approach exploits the correlations between firm characteristics and the likelihood of cyberattacks, and shows that the resulting estimate of cyber risk has predictive power for stock returns beyond the usual firm characteristics.

Finally, our paper is related to the emerging literature that applies computational linguistics to finance. This literature has made substantial progress in building subject matter lexicons. For instance, Tetlock (2007) used lexicons that are outside the finance field to detect tones in newspapers and their implications for returns and trading volumes. Loughran

data breaches.

[4] Kopp et al. (2017) and Warren et al. (2018) examine the impact of cyberattacks on financial institutions and financial market infrastructure, and argue such attacks potentially can be quite damaging. Duffie and Younger (2019) study the linkage between cyber security of large banks and financial stability. They argue that large banks tend to have sufficient liquidity to weather relatively extreme cyber runs; however, severe cyberattacks may deter nonbanks from sending funds through these institutions, which could create systemic risk.

[5] In untabulated results, we find a positive but moderate correlation between Florackis et al. (2022) and our cyber risk measure. In other words, the two approaches are consistent and complementary with each other.

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and McDonald (2011) developed a lexicon that classifies finance related words into different sentiment classes, and show that applying this lexicon to 10K filings can predict subsequent returns. More recently, researchers have explored deeper structures of texts that link to capital market activities. For example, Cohen et al. (2020) use changes in 10K texts to predict return, earnings, and bankruptcies. Ke et al. (2019) use supervised learning to extract sentiments from newspapers to predict returns. Bybee et al. (2020) apply topic modeling (an unsupervised word cluster approach) to gauge the relationship between news article topics and macroeconomic activities. Our paper contributes to the literature by introducing dictionaries developed in the cyber risk community to estimate individual firms' self-assessment of cyber risk in their 10K filings, and using it as an input to our machine learning algorithms to better predict cyberattacks.

The rest of the paper is organized as follows: Section 2 describes the data; Section 3 shows the methodology to construct the cyber risk measure; Section 4 presents the key results on the relationship between cyber risk and stock returns; Section 5 reports robustness tests; and Section 6 concludes.

2. Data

2.1. Cyberattacks

In this paper, we use a novel data set obtained from the Identity Theft Resource Center.[6] It has a number of advantages over the data set from the Privacy Rights Clearinghouse which has been used previously in the literature.[7] It provides more up-to-date data, reports many more cyberattacks on public firms, and importantly, contains the source of reports for the incidents, making it a more credible source. ITRC provides annual data breach reports from the year 2005, detailing all the reported and confirmed cyberattack incidents for US-based organizations.[8] These reports are stored on their website in the form of PDF files. We obtained the available annual reports from 2005 to 2019, and extracted all the items connected to cyber incidents. We first matched them to Compustat firms using a fuzzy matching algorithm, and then manually checked each one of the matched pairs for

[6] *ITRC*, `https://www.idtheftcenter.org/`. On their website: "The ITRC is a non-profit organization established to support victims of identity theft in resolving their cases, and to broaden public education and awareness in the understanding of identity theft, data breaches, cyber security, scams/fraud and privacy issues." See this SEC report that cites the data from ITRC: `https://www.sec.gov/files/speech-jackson-cybersecurity-2018-03-15-data-appendix-updated.pdf`.

[7] *Privacy Rights Clearinghouse*, 2019, `https://www.privacyrights.org/data-breaches`.

[8] There is a limitation however, as the data contains only "disclosed" incidents, since a firm has the discretion on whether to report incidents.

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confirmation. After this process, we are left with 1,010 cyberattack incidents.[9] Since we use the cyberattack as a binary variable, we count multiple attacks of a firm in a given year as one instance. This leaves us with 552 unique firm-year pairs, with 368 unique firms that experienced at least one cyberattack incident. Because we use many accounting variables as predictors, we follow the convention in Fama and French (1996), and define each year as the 12 months from July to the following June. The sample then spans 2006 to 2018.[10] Figure 1 plots the number of unique incidents and the unique firm-year pairs for firms that experienced cyberattacks during the sample period. It shows an increasing time trend in cyberattacks.

[Fig. 1 about here.]

An important question is whether the distribution across industries is uniform, as some industries are more digitized than others, and for some significantly more data exist. We plot the number of incidents for each of the Fama-French 17 industries per Fama and French (1988) in our sample in Figure 2.

[Fig. 2 about here.]

From Figure 2, we find that the financial sector is the hardest-hit industry, with a total of 247 incidents, dwarfing all other sectors. This is not surprising because financial firms possess large amounts of client identification and financial data. With the exception of "Mining and Minerals" industry, no sector appears immune to cyberattacks as firms increasingly rely on online platforms to conduct their business and on cloud services to store their data.

2.2. Other Data

The rest of the data sources are as follows. Stock price and return data are from the CRSP. Accounting data are from the Compustat. Asset pricing factors are from Kenneth French's website.[11] The 10-K filings are from the University of Notre Dame Repository website.[12]

[9] This increases our data set from the 311 observations from the PRC, which is nearly 3-fold increase.

[10] We drop the first 6 months of the calendar year 2006 and the last 6 months of the year 2019, as they do not constitute a whole year per our definition.

[11] *French, Kenneth*, 2019, `http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`.

[12] *McDonald, Bill, University of Notre Dame*, `https://sraf.nd.edu/data/stage-one-10-x-parse-data/`.

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2.3. Summary Statistics

Table 1 summarizes the characteristics of firms that experience cyberattacks (victims) and those that do not (non-victims) in our sample. All the variables are measured prior to the year when the incidents happen. The accounting variables are lagged by six months to ensure data availability. We follow Kamiya et al. (2021) in the definitions of the firm characteristics. In addition, we use two linguistic variables: the "Risk Factor Length," which is based on the length of "Item 1A: Risk Factors" in firm's 10-K disclosure, and the "NIST Counts," which is the number of times a firm mentions a cybersecurity related term in Item 1A, based on the NIST dictionary. We discuss this dictionary in more detail in Section 3.

[Table 1 about here.]

Table 1 shows that the victim firms tend to be larger, which suggests that hackers tend to target firms with larger customer bases and bigger data sets, as such targets are likely to offer them potentially higher gains. They tend to be have more intangible assets, have higher cash flow, spend less on research and development, have higher return on assets, and experience higher past year excess returns. These characteristic differences are in line with the results in Kamiya et al. (2021). In addition, victims tend to spend more time discussing cybersecurity related risks in their 10-K disclosure. In the next section, we describe how we use these characteristics to construct the firm-level measure of cyber risk.

3. Measuring Firm-Level Cyber Risk

A well-known insurer, the Northbridge Insurance, provides a broad definition of cyber risk: "Cyber risk commonly refers to any risk of financial loss, disruption or damage to the reputation of an organization resulting from the failure of its information technology systems."[13] To operationalize this notion, we follow the literature (e.g., Kamiya et al. (2021) and Michel et al. (2020)), and treat reported cyberattacks as realized cyber risk. Based on these cyberattacks, we use machine learning algorithms to develop a real-time estimate of the likelihood for each individual firm to experience a cyberattack in the subsequent period.

3.1. Constructing Predictors

Kamiya et al. (2021) document that a set of firm specific variables are correlated with the probability for a firm to be a victim of a cyberattack. We use these variables as the

[13] *Northbridge Insurance*, 2019, `https://www.nbins.com/blog/cyber-risk/what-is-cyber-risk-2/`.

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starting point in building our model. The variables include the asset intangibility, the ratio of capital expenditures to total assets (CAPX/AT), the ratio of cash flows to total assets (CF/AT), firm size, net worth, R&D, ROA, Tobin's q, sales growth, financial constraint, prior year excess return over the CRSP value-weighted market return, and the natural log of firm age. In addition, we explore soft information about a firm's cyber vulnerability. One source of such soft information is a firm's self-assessment of its cyber risk disclosed in "Item 1A: Risk Factors" section in firms' annual 10-K filings. The SEC requires that firms disclose the most significant risk factors.[14] Per this requirement, firms started to include this section prominently and regularly from 2006. Firms are in a unique position to assess their own risk level given their private information and familiarity with the firm's operations. In addition, this risk disclosure is forward-looking, providing useful information about the perceived exposures. The nature and intensity of the discussion on cyber related risk in their 10-K filings should be a useful indicator of their firms' vulnerability to cyberattacks.

To measure the intensity of perceived cyber risk, we count the number of times a firm mentions cyber risk-related terms in the risk factors section in its 10K filing as well as the length of the entire section. In this way, we can capture the information about a firms' self assessment of both its overall risk exposure and exposure specific to cyber risk.

To identify cyber risk terms, we exploit a specialized dictionary, compiled by the National Institute of Standards and Technology (NIST), named the "Glossary of Key Information Security Terms."[15] An important feature worth noting is that the NIST dictionary was compiled in 2006, suggesting that cyber risk was already a concern by governmental bodies. Secondly, it avoids a potential look-ahead bias since our sample starts in the year 2006.[16]

Next, we turn to firms' annual 10-K filings. Since the SEC mandates of disclosing risk factors came into effect in 2006, we start our sample in 2006. We obtain the filings from 2006 to 2018 through the University of Notre Dame Repository, and extract their risk factor sections.[17] We use the central index key (CIK) to match the filing firms to Compustat firms. Then we count the number of tokens per risk factor text block, only keeping the blocks with at least 100 tokens.[18]

The final step in constructing the cyber risk disclosure variable is to measure the intensity

[14] See *SEC*, 2005, `https://www.sec.gov/rules/final/33-8591.pdf`.

[15] *NIST*, April 25, 2006, `https://www.nist.gov/publications/glossary-key-information-security-terms`.

[16] A contemporaneous study by Jamilov et al. (2020) uses a different set of words and phrases as a dictionary. We show in Section 5 that our results are robust to using alternative dictionaries.

[17] We use an algorithm to match the section between the beginning of Section 1A and either Section 1B or Section 2.

[18] A token is an instance of a sequence of characters in some particular document that are grouped together as a useful semantic unit for processing (Schütze et al. (2008)). Here we refer to each term as a token, after preprocessing the text such as dropping stop words.

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with which cybersecurity related terms are mentioned. We apply the following procedure: first, we measure the maximum number of tokens in each term in each dictionary. We find that the NIST dictionary contains terms that can be as long as seven tokens. Second, to control for the fact that there might be common boilerplate language that most firms use, we eliminate the common terms used by firms in each year as in Hoberg and Philips (2016). Third, we transform each risk factor text block into a collection of $Ngram$s, where $N \in [1, 2, 3, 4, 5, 6, 7]$. Each $Ngram$ is a collection of every possible combination of adjacent N tokens in each document.[19] Finally we count how many times each cybersecurity term occurs in each $Ngram$-transformed text block.

We use the length of the risk factor section and the NIST term frequency as two separate variables for two reasons. First, the length of risk factor section measures the number of tokens, or uni-grams, while the NIST term frequency the number of times each term is mentioned. Here the term ranges from uni-grams to seven-grams, and thus it is inappropriate to scale the frequency by the length of the section. Second, we keep both variables in the feature set to maximize the power of the machine learning techniques.

3.2. Constructing the Predictive Model

3.2.1. In-sample Fit

Before building the predictive models, we examine the in-sample explanatory power of the regressors we select in-sample. In particular, we perform in-sample logit regressions which include the accounting and text-based disclosure variables, which are cross-sectionally standardized to have means of zero and standard deviations of one. Following Petersen (2009), we cluster standard errors by both firm and year.

The results in Table A.1 of the Appendix show that many firm characteristics have strong associations with the probability of future cyberattacks. In terms of magnitudes, firm size is particularly important, with larger firms more likely to be attacked. We also find that the text-based cyber risk count variables have positive predictive power for future cyberattacks. Interestingly, as shown in Column (1), the total number of tokens in the Risk Factors section of a firm's 10-K filing has a negative association with the probability of future cyberattacks. This result suggests that firms with more thorough discussions of their risk factors may also have better risk management practices.

[19] See, e.g., Damashek (1995) for description of Ngrams.

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3.2.2. Building Machine Learning Predicting Models

To build an effective cyberattack forecasting model, there are two main considerations. First, an in-sample fit could induce the look-ahead bias and overfitting. Second, cyberattacks are rare events, which results in a highly imbalanced sample, thereby introducing biases into the Maximum Likelihood Estimator (King and Zeng (2001)).

To address the possible look-ahead bias, we follow a recursive and expanding prediction procedure. Namely, we start from the year 2006, run a predictive model, save the coefficients, and then use them to fit the data in 2007. We do this recursively, so the final step is to use the data from 2006 to 2017 to train the model to fit the year 2018. In this way, we avoid the look-ahead bias, and generate true out-of-sample (OOS) predictions.

Next, to address overfitting and maximize OOS performance, we introduce machine learning classification models to improve the predictive power. In particular, we select logistic ridge regression as our main model.[20] The logistic ridge regression combines the logistic regression with ridge regularization, which uses an L-2 penalty.[21] The objective function for logistic ridge regression is:

$$\min_{\beta_0,\boldsymbol{\beta}\in\mathbb{R}^{p+1}} -\Big[\frac{1}{N}\sum_{i=1}^{N} y_i \cdot (\beta_0 + \boldsymbol{x_i}^T\boldsymbol{\beta}) - \log(1 + e^{\beta_0+\boldsymbol{x_i}^T\boldsymbol{\beta}})\Big] + \frac{1}{2}\lambda\|\boldsymbol{\beta}\|_2^2 \tag{1}$$

where p is the number of parameters, and λ controls the regularization strength.

To fully exploit the conditioning information and the ridge regression's regularization capability, we perform the polynomial transformation of our regressors.[22] A polynomial transformation of degree d converts the variables to all possible interactions and powers up to degree d. In our implementation, we choose a polynomial transformation of degree 4.

In each training window, we tune hyper parameters for optimal performance via three-fold cross-validation, using the training data in each recursive sample period. Cross-validation is done through the following procedure: for the training data in each window, we randomly split the training data into three folds, i.e., three subsets of equal size; we use two of the three folds to fit the model, and the one remaining fold as the "validation set" to find the best estimator; we iterate this procedure three times and find the best estimator, in terms of out-of-sample performance metric (based on the validation set of each iteration); then we

[20] We show in Section 5 additional results based on Ensemble methods, and find our results to be robust to model choice.

[21] Another popular algorithm in asset pricing is LASSO, which uses an L-1 penalty to achieve model sparsity. Since we are primarily interested in the forecasting performance of the model rather than variable selection, we focus on the ridge regression. With LASSO, we are able to report weaker but consistent results.

[22] See Hastie et al. (2017) for an introduction of polynomial transformation. This function is implicitly used in kernel methods such as support vector machines.

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use this estimator for prediction on the test data. Since regularization is sensitive to the unit of variables, we scale the variables before the training process (Hastie et al., 2017).

Since our sample is highly imbalanced, we use the stratified K-fold strategy of Zeng and Martinez (2000), to ensure that the class distribution is maintained across all folds. In addition to tuning the penalty factor λ, we also tune the class weight parameter in the loss function, using a heuristic proposed by King and Zeng (2001).[23]

3.2.3. Filtering the Sample

In our study, we wish to capture significant cyberattacks with meaningful impacts on the victim companies. In a contemporaneous study, Florackis et al. (2022) select events that are prominently featured in media reports as a filter that identifies important attacks. We take a similar but more direct approach: we use a filter based on the 7-day cumulative abnormal returns (CAR) around the events. Specifically, we first calculate the [-3,3] window CAR for each event using a four-factor model including the Fama-French three factors and the momentum factor as the benchmark. Then we choose the events with an absolute value of CAR greater than 1%; we consider these events as important to investors based on the stronger market reaction.[24] After using this filter, our sample consists of 394 cyberattacks ranging from 2007 to 2018.

3.2.4. Evaluating the Forecasting Performance

To evaluate the performance of the logistic ridge regression against the simple logistic regression, we follow the machine learning literature and choose three commonly used metrics of forecasting performance: the F1-score, G-mean, and Balanced Accuracy.[25]

The building blocks for these metrics include recall (also called sensitivity), precision, and specificity.

$$Recall = Sensitivity = \frac{True\,Positives}{True\,Positives + False\,Negatives}$$

[23] King and Zeng (2001) shows that in highly imbalanced learning problems, the coefficients can be biased towards the majority class, effectively rendering the model useless. They propose several measures to mitigate the bias, including the prior correction and weighted logistic regression. We use the second option as it is less costly and shows similar performance compared with the prior correction procedure. The weighted logit equation is: $\ln \mathcal{L}_w(\beta|y) = w_1 \sum_{Y_i=1} \ln \pi_i + w_0 \sum_{Y_i=0} \ln (1-\pi_i)$. The intuition for the weighted logistic is that the objective function is weighted by a coefficient w that is inversely related to the sample class distribution. Consequently the minority class would be weighted more heavily and thus mitigate the bias. We tune w via cross-validation to add extra flexibility.

[24] We choose 1% because Kamiya et al. (2021) document that attacked firms on average experienced -1% CAR around the events. Our results are robust to different specifications. We show in Section 5 that using 0.5% absolute CAR as the filter or not using any filter produces similar results.

[25] See Brodersen et al. (2010); Yue et al. (2007); He and Garcia (2009) for discussions of these metrics.

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$$Precision = \frac{True\,Positives}{True\,Positives + False\,Positives}$$

$$Specificity = \frac{True\,Negatives}{True\,Negatives + False\,Positives}$$

F1 score is the harmonic average of the recall and precision; G-mean is the geometric average of the recall and specificity; and Balanced Accuracy is the arithmetic average of sensitivity (recall) and specificity.

Table 2 compares the average forecasting performance of the logistic ridge regression with that of the commonly used simple logistic regression over our sample period. Figure 3 presents the time-series of the performance metrics.

[Table 2 about here.]

[Fig. 3 about here.]

The results show that the logistic ridge regression strongly outperforms the baseline logistic regression across all three performance metrics. For instance, the average F1 score of the logistic ridge regression is approximately 15 times as large as that of the logistic regression, and the average G-mean of the logistic ridge regression is approximately 40 times as large as that of the logistic regression. The Balanced Accuracy of the logistic ridge regression exceeds that of the logistic regression by approximately 30%. The strong outperformance of the logistic ridge regression is persistent through time.

To understand the drivers of the superior performance of the logistic ridge regression, we plot the aggregate (or equivalently, mean) confusion matrices for the logistic ridge and logistic regressions in Figure 4. The rows of a confusion matrix are the true classes, while the columns are predicted classes. Each of the four quadrants indicates the total number of observations across the sample period classified per the prediction model.

[Fig. 4 about here.]

In a confusion matrix, a superior forecasting technique tends to register higher numbers in the diagonal elements, which imply that the classifier tends to more correctly classify observations. The results show the source of the weakness of the simple logistic regression: it overwhelmingly predicts firms as negative cases and under-predicts the positive cases. Among the firms that experience a cyberattack the following year, the algorithm falsely predicts more than 99% of them as negative cases. In comparison, the logistic ridge regression falsely predicts approximately 50% of them as negative cases. In other words, the logistic

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ridge regression benefits from a jump in recall in exchange of a small decrease in precision, and thus achieves a better balance in performance.

In the rest of the paper, we use the cyber risk measure based on the ridge regression as the primary measure. We present the results based on alternative machine learning techniques in Section 5 as robustness tests.

3.2.5. Characterizing the Firms with High Cyber Risk

Figure 5 shows the correlation coefficients between our cyber risk measure and firm characteristics. The left column presents the time-series averages of the cross-sectional Spearman correlations and the heat map is based on the absolute value of the average correlation coefficient.

[Fig. 5 about here.]

The results indicate that firms with higher cyber risk tend to be larger, less financially constrained and more profitable on the basis of the average correlation coefficients. The heat map also shows that these three characteristics tend to have high and stable correlations with cyber risk across time. In addition, the text-based variables have high correlations with cyber risk.

4. Cyber Risk and Stock Returns

In this section, we study the relationship between cyber risk and stock returns. We start with cross-sectional analyses using both the Fama and MacBeth (1973) regressions and portfolio analyses. Then we examine the time variation in the return spread between firms with high and low cyber risk, focusing on its comovement with other measures of aggregate cyber risk concerns.

4.1. Cross-sectional Regressions

To examine whether firms with higher cyber risk compensate investors with higher average returns, we use the Fama-MacBeth cross-sectional regressions to examine the incremental predictive power of the cyber risk measure for stock returns, controlling for well-known stock characteristics proposed in the previous literature. Specifically, at the end of each month from July of year t to June of year $t+1$, we regress individual stocks returns on our cyber risk measure estimated in year t and a set of firm characteristics. The cyber risk estimate is based on firms' accounting information for the fiscal year ending anytime in the calendar

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year $t-1$ and the cyberattack forecasting model parameters estimated using the accounting information available up to June in year $t-2$, and the cyberattack incidents available up the end of June of year $t-1$. The year t ranges from 2008 to 2018. All the regressors are cross-sectionally standardized to have a mean of zero and standard deviation of one. Then we test whether the time-series averages of the regression coefficients are statistically significant. The standard errors are based on the Newey and West (1987) adjustment with 6 lags. For a stock to be included into the analysis, it is required to have a CRSP share code 10 or 11, and to have a closing price above \$5 at the end of June in year t.

Table 3 shows the results. In Column (1), we include the natural log of market cap, natural log of book-to-market ratio, gross profitability, asset growth, and momentum (measured as prior twelve to one-month return). Column (2) further includes the past onep-month return to control for the short-term return reversal, idiosyncratic volatility (Ang et al., 2006), and illiquidity (Amihud, 2002). Column (3) adds the organizational capital (Eisfeldt and Papanikolaou, 2013), CAPM beta, tail risk beta (Kelly and Jiang, 2014), coskewness (Harvey and Siddique, 2000), and net operating assets (Hirshleifer et al., 2004).

[Table 3 about here.]

In these regressions, the coefficients for the cyber risk measure are economically meaningful and statistically significant. For instance, Columns (1) shows that a one-standard-deviation increase in cyber risk is associated with an approximately 20 basis-point increase in returns per month, or 2.4% per year. The effect is statistically significant at the 1% level. The positive relation between cyber risk and future stock returns remains strong after controlling for other firm characteristics in Columns (2) and (3). These results show that cyber risk is an important driver of the cross-sectional variation in average stock returns in our sample period.

4.2. Portfolio Sorts

We also use the portfolio approach to examine the relation between cyber risk and stock returns. At the end of each June from 2008 to 2018, we sort stocks into three tercile portfolios, four quartile portfolios, or five quintile portfolios according to the cyber risk measure. These portfolio are held until the end of June in year $t+1$ and then rebalanced based on the updated cyber risk estimate when new information is available.

We create zero-cost hedge portfolios that buy high cyber risk stocks and short low cyber risk stocks. We compute the monthly equal- and value-weighted alpha on the hedge portfolios using a number of asset pricing models: the CAPM, Fama-French three-factor model (FF3),

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FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6).

Table 4 shows that the hedge portfolios generate large alphas across the different portfolio constructions and against the various benchmark models. The monthly alpha ranges from 40 to 80 basis points per month and is always statistically significant. These results show that investors in high cyber risk stocks earn high average returns, which cannot be explained by these asset pricing models.

[Table 4 about here.]

4.3. Comovement with the Index of Cybersecurity

In addition to the cross-sectional variation in average returns, we examine the time variation in the relative returns between stocks with high and low cyber risk. In this subsection, we relate the return spread to an independent survey conducted by the New York University on the perception of cyber risk among industry experts: the Index of Cybersecurity.[26] Each month, the survey queries experts such as chief risk officers, chief information security officers, selected academicians engaged in field work, and selected security product vendors' chief scientists about their perceived level of cyber risk, from which it builds an aggregate measure of cybersecurity.

To examine the relation between our annual firm-level cyber risk measure and the the monthly aggregate ICS, we project both variables onto the return space. For each stock in our sample, we estimate its return sensitivity to the monthly percentage change in the ICS. That is, we perform bivariate regressions of monthly excess returns of individual stocks on the monthly changes in the ICS and the excess returns on the market portfolio. Stocks with a high (low) ICS beta tend to have lower (higher) cyber risk exposures based on the ICS. Therefore, a long-short portfolio that buys stocks with a high ICS beta and sells those with a low ICS beta should deliver higher performance when the aggregate cyber risk concern is high. If our firm-level cyber risk measure and the ICS capture some common component of cyber risk in the economy, we would expect the portfolio that buys high cyber risk stocks and sells low cyber risk stocks to have a negative correlation with the ICS factor-mimicking portfolio.

We apply a 12-month rolling window to estimate the ICS beta for individual stocks, requiring at least six observations for the statistical estimation. We form five quintile portfolios at the end of each month from December 2015. Then we repeat this process for each subse-

[26] ICS, *NYU Engineering*, `https://wp.nyu.edu/awm1/`.

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quent month. Therefore, we have monthly ICS beta portfolios from December 2015 to June 2019, where we have concurrent observations for our cyber risk based quintile portfolios.

Panel A of Figure 6 shows the performance of the portfolio that buys stocks with high cyber risk and sells those with low cyber risk (the solid black line) against the return on the portfolio that buys high ICS beta stocks and sells low ICS beta stocks (the dotted red line). Consistent with our conjecture, the two series show a strong negative comovement. The time-series correlation coefficient is -41.2% and statistically significant.

4.4. Comovement with the Cybersecurity ETFs

In this subsection, we use the return on cybersecurity index ETFs as another proxy for the aggregate concern about cyber risk. The idea is that when investors have stronger concerns about cyber risk, the cybersecurity ETFs tend to outperform. Our analyses include two such ETFs: the First Trust Nasdaq Cybersecurity ETF (CIBR)[27] and ETFMG Prime Cyber Security ETF (HACK).[28] Both ETFs track a set of firms that provide cybersecurity services. Table 5 shows their top ten holdings.

[Table 5 about here.]

Panel B of Figure 6 plots the monthly returns on the spread portfolio that buys high cyber risk stocks and sells low cyber risk stocks (the solid dark line) against the performance of the cybersecurity ETFs (the dotted blue and dashed green lines). It shows a strong negative comovement. The time-series correlations are below -40% for both ETFs. If we perform a time series regression of our cyber risk portfolio return against the two ETF returns individually, the coefficients are -0.405 and -0.312 respectively for CIBR and HACK, and are statistically significant at the 1% level. This result provides further validation for our cyber risk measure.

[Fig. 6 about here.]

5. Robustness Tests

5.1. Sub-sample Analysis and Industry Effects

To evaluate the robustness of our results, we split our sample into two subperiods and perform the Fama-MacBeth regressions of stock returns on cyber risk and other characteristics for each subperiod. The first two columns of Table 6 show the results. We find that in

[27] CIBR, `https://www.ftportfolios.com/retail/etf/etfsummary.aspx?Ticker=CIBR`.
[28] HACK, `https://etfmg.com/funds/hack/`.

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both periods, the cyber risk measure is positively related to future stock returns, with the effect slightly stronger in the more recent subperiod.

It is also possible that the firms with high cyber risk based on our measure may over-represent the firms from industries that benefit more from digitization, which may in turn drive their relatively high performance. To address this concern, for each stock in our sample, we subtract the industry portfolio return from the stock's monthly returns based on the Fama-French 48 industry classifications (Fama and French, 1997), to arrive at the industry-adjusted excess returns. We then perform the Fama-MacBeth cross-sectional regressions with the industry-adjusted return as the dependent variable and our cyber risk measure and other firm characteristics as independent variables. We report the coefficients in Column (3) of Table 6. The results show that even after controlling for industry returns, cyber risk is positively and significantly correlated with future stock returns.

[Table 6 about here.]

5.2. Alternative Machine Learning Models

In this subsection, we present the results using alternative machine learning models to estimate the ex-ante cyber risk. In particular, we follow Liu et al. (2008), using a combination of undersampling and bootstrapping in an Ensemble method, termed "EasyEnsemble" (EEC).

The procedure of this method is as follows. In each training window, we randomly sample a subset of non-victim observations and pair them with the victim set, which ensures an equal number of non-victim and victim firms, thereby constructing a "balanced" sample. We fit an estimator on this sample and save the parameters. We repeat this process for 50 times by independently sampling 50 subsets from the non-victim set. Based on these bootstrapped samples and results, we obtain an ensemble and arrive at a final estimate. Liu et al. (2008) argues that this procedure has superior forecasting performance.

In our setting, we present two additional sets of results using K-nearest neighbors algorithm and Naive Bayes algorithm as the base estimators for EEC. We find that the cyber risk measures using the alternative machine learning models also have superior out-of-sample forecasting performance. In Figure 7, We plot the time series of the forecasting performance metrics for the measures based on the logistic regression, logistic ridge regression, EEC-KNN, and EEC-NB techniques. It shows that all the three machine learning techniques outperform the simple logistic regression.

[Fig. 7 about here.]

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Moving to the asset pricing tests, we find that the cyber risk measure based on alternative machine learning techniques also show a strong relation to future stock returns. For instance, Table 7 presents the portfolio performance based on the alternative cyber risk measure. The alpha on the long-short portfolio remains large and statistically significant.

[Table 7 about here.]

5.3. Placebo Test

To address the concern that the relation between our cyber risk measure and future stock returns is driven merely by the correlations between the returns and characteristics of the firms that were exposed to cyberattacks, we design the following placebo test. For each hacked firm in each year, we randomly select a firm from the same Fama-French 48 industry, and label it as the "pseudo" attacked firm. We repeat this for all the attacked firms in our sample, and thus create a sample of 394 pseudo attacked firms. With this pseudo sample of hacked firms, we use the same statistical procedure as in our main analyses to estimate a cyber risk measure for each firm. We then test whether the resulting pseudo cyber risk measure (a mechanical combination of stock characteristics) shows a reliable relation to future stock returns. The results from our portfolio tests, as shown in in Table 8, indicate that there is no reliable relation between this pseudo cyber risk measure and future stock returns.

[Table 8 about here.]

5.4. Alternative Dictionary

In a contemporaneous study, Jamilov et al. (2020) use a set of 30 cyber risk related words and phrases to capture the level of cyber-related concerns expressed by corporate executives express in their quarterly earnings conference calls. We evaluate the robustness of our results using this alternative dictionary.

Specifically, we count the number of times a firm uses these cyber related words and phrases in the Item 1A "Risk Factors" section of the 10-K filings. We replace our original text variable constructed based on the NIST glossary with the new variable. Then we apply the same statistical techniques to estimate our cyber risk measures. The results in the Appendix show a robust relation between these cyber risk measures and future stock returns.

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6. Conclusion

In this paper, we use the machine learning algorithms to develop an ex-ante cyber risk measure for individual firms, which has superior ability to forecast the occurrence of future cyberattacks. We find that firms with higher cyber risk according to this measure earn higher average stock returns, which cannot be explained by standard asset pricing models. In times when these firms underperform, cybersecurity experts tend to have higher concerns about cyber risk and the cybersecurity exchange-traded funds outperform. These results suggest that cyber risk may be an important determinant of expected returns in the increasingly digitized economies.

Our study suggests interesting avenues for future research. First, we have followed a bottom-up approach to estimate the cyber risk premium starting with firm-level estimates of cyber risk. Another approach, which is top-down, can examine the systemic impact of a large attack on major players in the economy. For instance, Eisenbach et al. (2020) model how cyber attacks on large banks can influence the financial system and even the economy through network effects. It would be of interest to explore the connections between these two broad approaches.

Second, a limitation of our study is that we focus on the likelihood of cyberattacks as a proxy for cyber risk, but do not study the scale of the loss resulting from cyberattacks. Although the probability of cyberattacks is likely to be correlated with the severity of cyberattacks, these two dimensions of cyber risk can contain different information. Our research makes a useful first step towards understanding the implications of cyber risk for asset prices. Future research can benefit from investigating these two dimensions in a unified framework.

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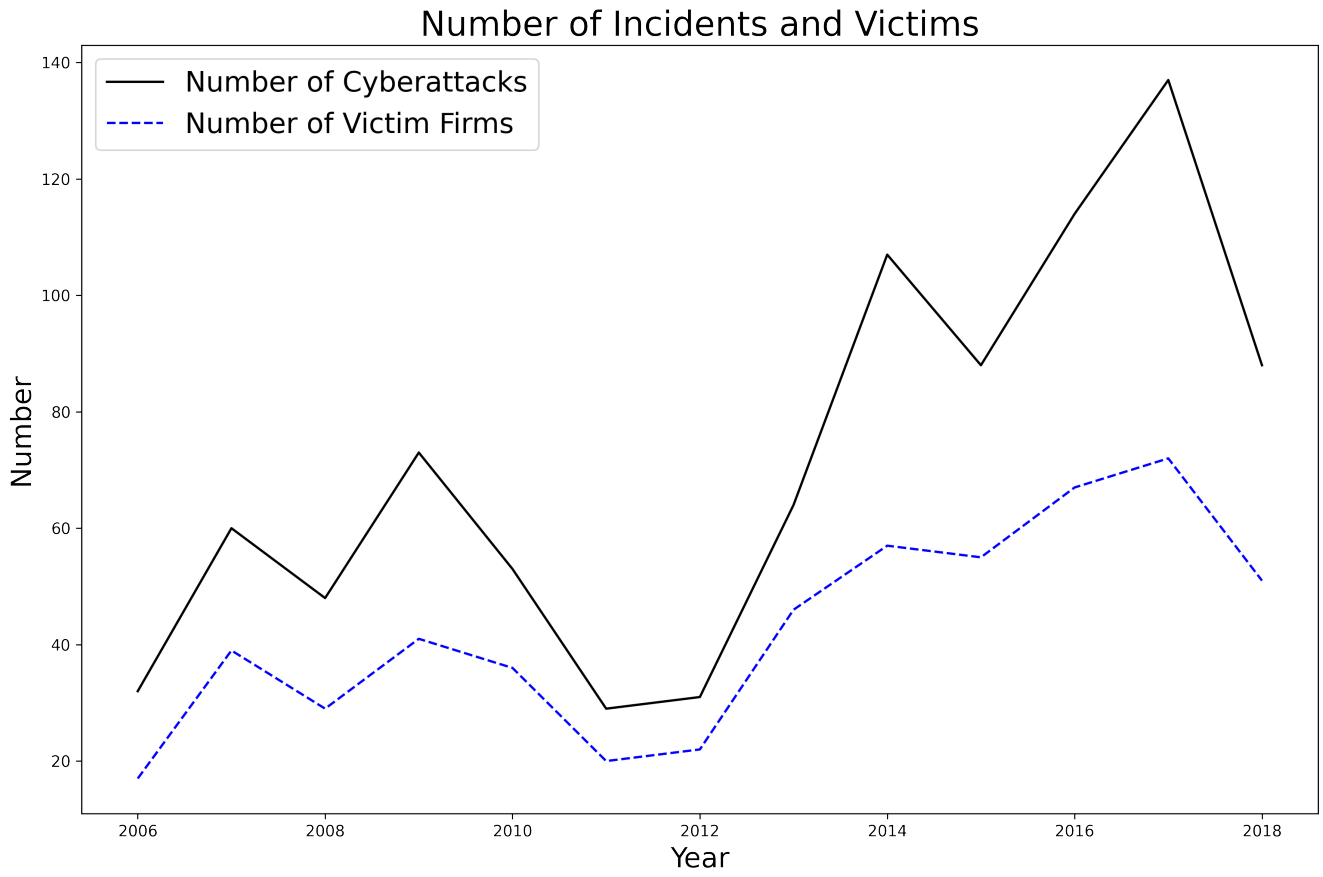


Fig. 1. Disclosed Cyberattack Incidents for US Public Firms. The figure depicts the number of cyberattack incidents that are disclosed to have occurred to US public firms from year 2006 to 2018. The year convention follows Fama and French (1996), which starts from July to next June. Source of data is Identity Theft Resource Center (ITRC). The black line indicates the number of unique cyberattack incidents matched to Compustat firms; the blue dashed line shows the number of unique public firms that experienced cyberattack incidents each year.

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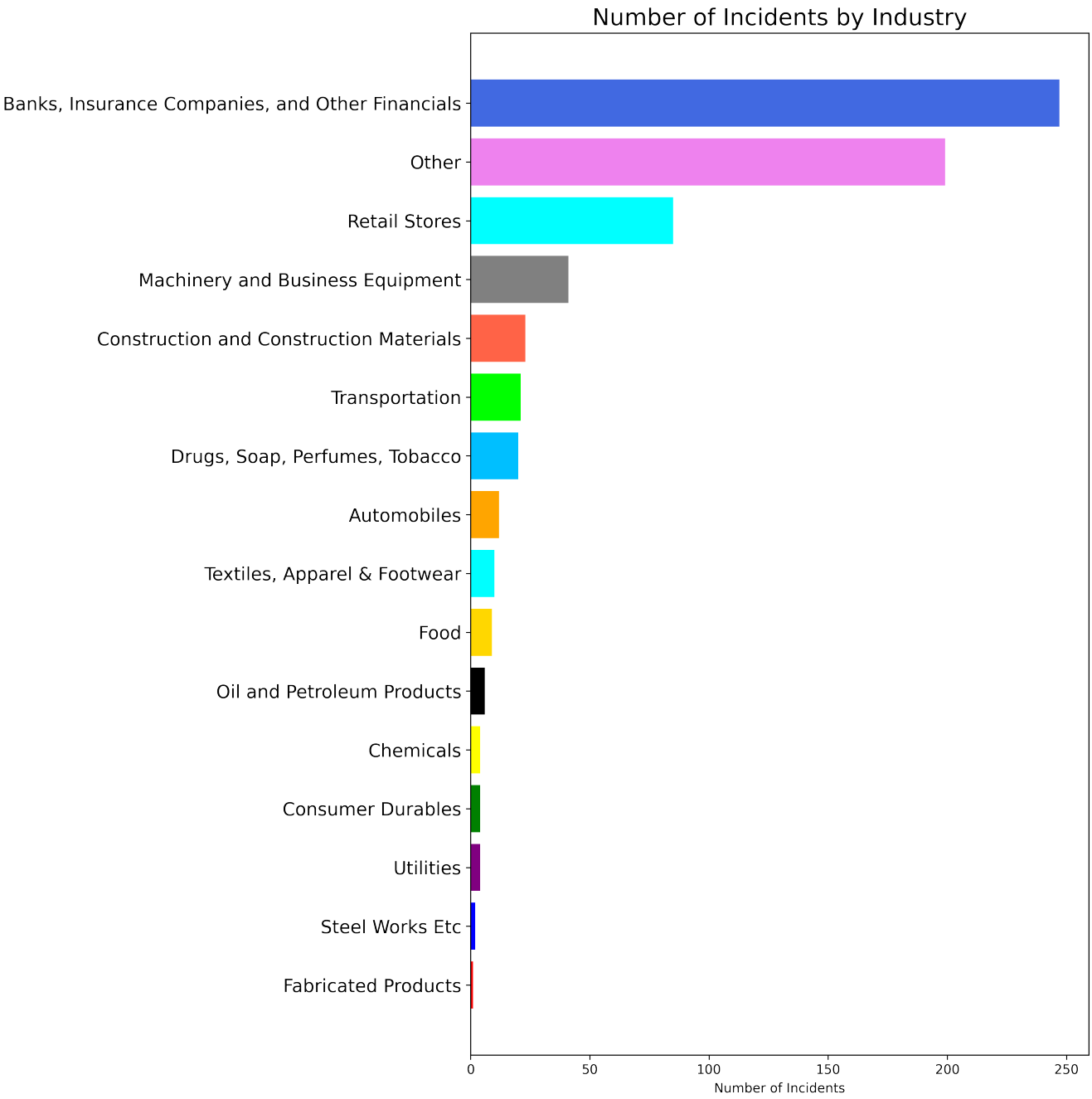


Fig. 2. Industry Distribution of Cyberattacks. The figure depicts the industry distribution of attacked firms based on Fama-French 17 industry classification per Fama and French (1988). The bars show the number of incidents in the sample period in each industry. The hardest hit is the financial industry, with a total of 247 incidents. The least represented is the "Mining and Minerals" industry, which has zero incident in our sample. The sample runs from 2006 to 2018.

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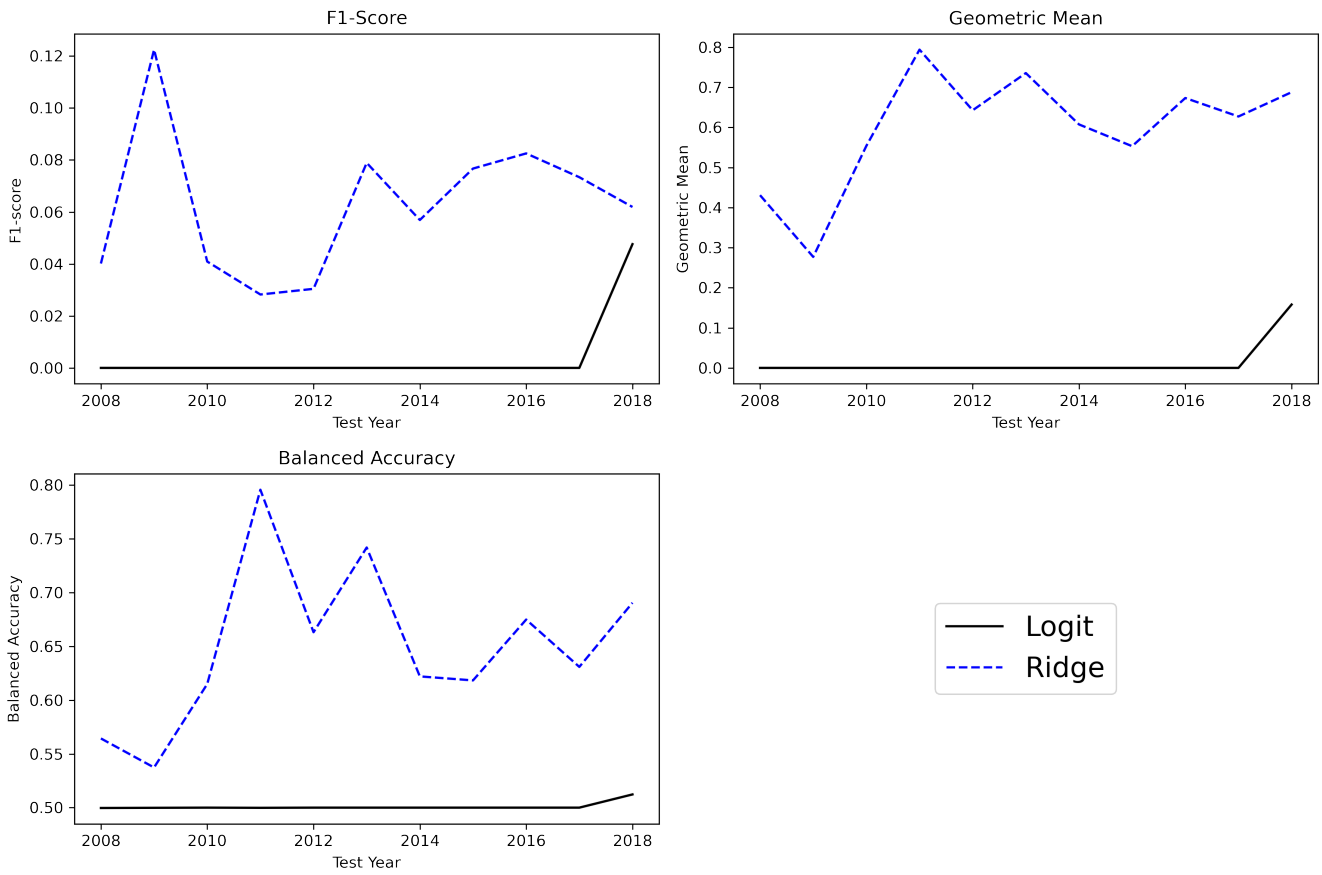


Fig. 3. Out-of-Sample Performance Metrics. The figure depicts the time series of three out-of-sample performance metrics for both logit and ridge regression across the test years. The three metrics are: F1 score, G-mean, and Balanced Accuracy. F1-score is the harmonic mean between precision and recall. Geometric mean is the geometric mean between sensitivity and specificity. Balanced accuracy is the arthmetic mean between sensitivity and specificity. The black solid line indicates logit, while the blue dashed line represents ridge. The metrics are all measured against test samples, and thus the figure runs from 2008 to 2018.

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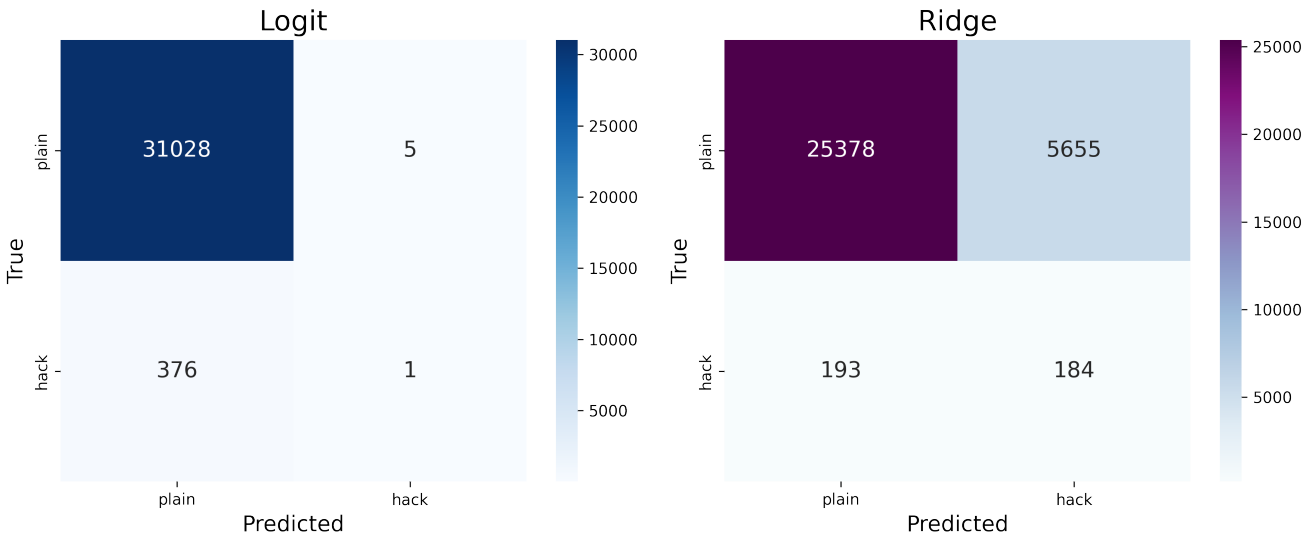


Fig. 4. Confusion Matrices. The figure shows the aggregate/mean confusion matrices for logit and ridge regression. That is, we aggregate the numbers in the four quadrants across the 11 test windows for each model. The rows are true classes, while the columns are predicted classes. Each quadrant presents the number of observations that the classification model allocates. The diagonal numbers represent the total number of observations that are correctly allocated to their respective classes. For example, for logit in the left panel, it correctly predicts 1 cyberattack case, while mis-allocateing 376 cases as non-victims. On the other hand, ridge regression in the right panel correctly predicts 184 cyberattack cases, while mis-allocating 193 cases in the non-victim class.

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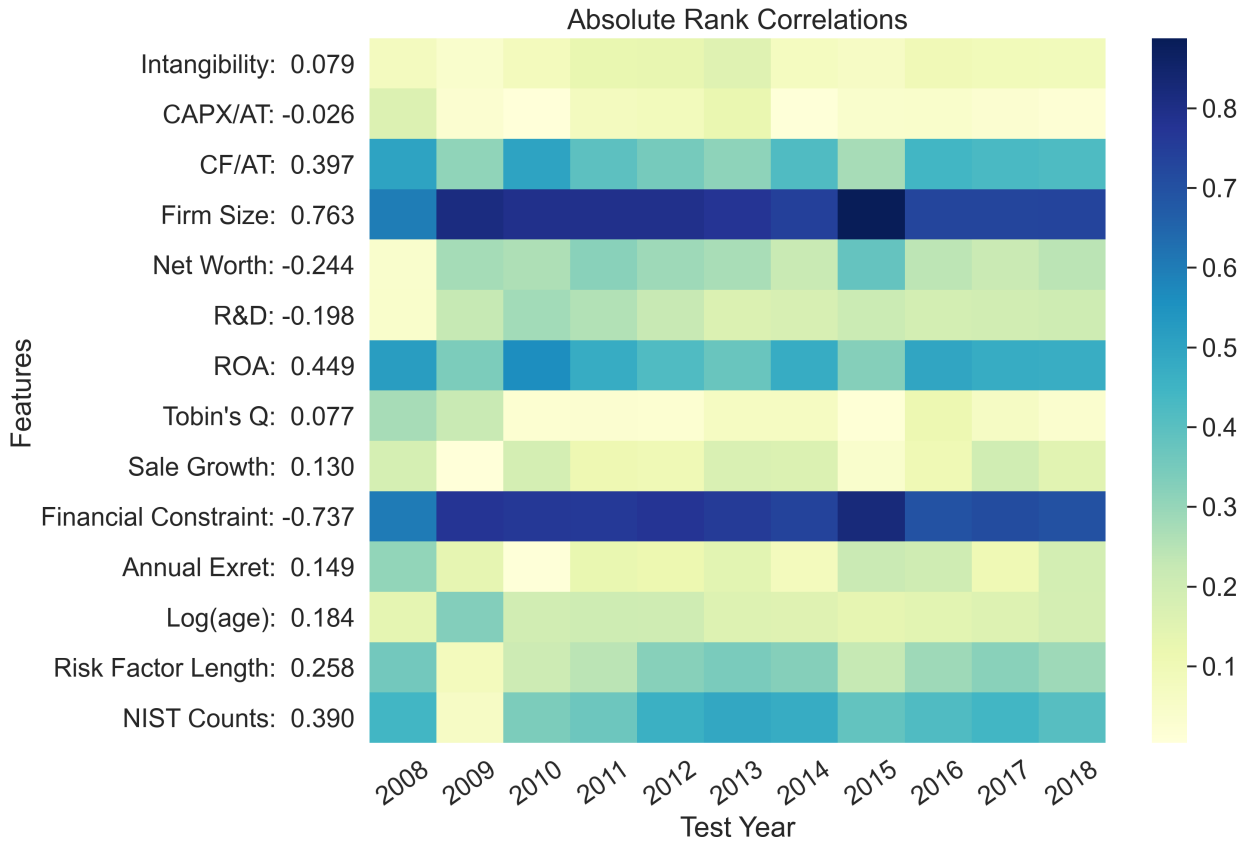


Fig. 5. Absolute Rank Correlations between Predictors and Cyber Risk. The figure depicts the absolute value of rank correlations between each of the 14 predictor variables and the estimated cyber risk using ridge regression for each test year, as well as the time-series mean of rank correlations for each variable. Specifically, for each test year, we compute the rank correlation between each variable and cyber risk measure, and then take time-series mean for each variable, and put them in the left column of the figure. Then we compute the absolute values of all the rank correlations of all variables across the test years. In so doing, we have 11 sets of coefficients, since the test years run from 2008 to 2018. We plot them in a heat map as shown in the figure. The color depth represents how large the coefficient is relative to other variables. For example, firm size is consistently important across years, while CAPX/AT has been consistently less important.

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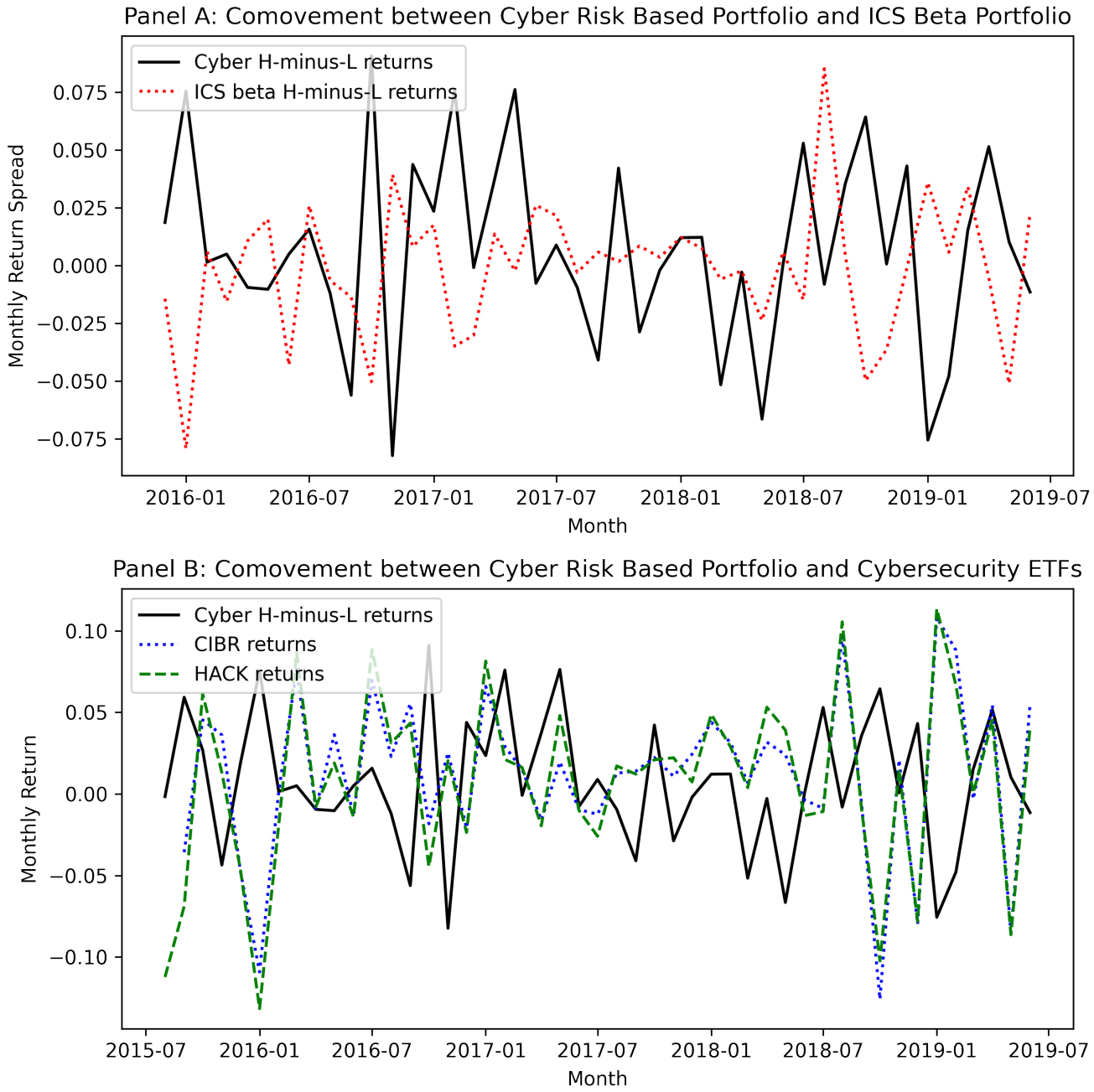


Fig. 6. Comovement with ICS and Cybersecurity ETFs. The figure shows the comovement of the return series of our high-minus-low cyber risk portfolio and that of ICS beta portfolio and that of two cybersecurity ETFs. In Panel A, we plot the return series of our high-minus-low cyber risk portfolio and ICS beta portfolio. The ICS beta portfolio is constructed at a monthly frequency, by sorting stocks into quintile portfolios, based on stocks' return betas on the monthly change of ICS index, controlling for market excess returns during the past 12 months. The two series have a correlation of -41.2%. In Panel B, we plot the monthly returns of the First Trust Nasdaq Cybersecurity ETF (CIBR) and ETFMG Prime Cyber Security ETF (HACK), from August 2015 to June 2019. The time-series correlation between the returns on the long-short cyber risk portfolio and the ETFs are -47.6% and -41.5% for CIBR and HACK, respectively.

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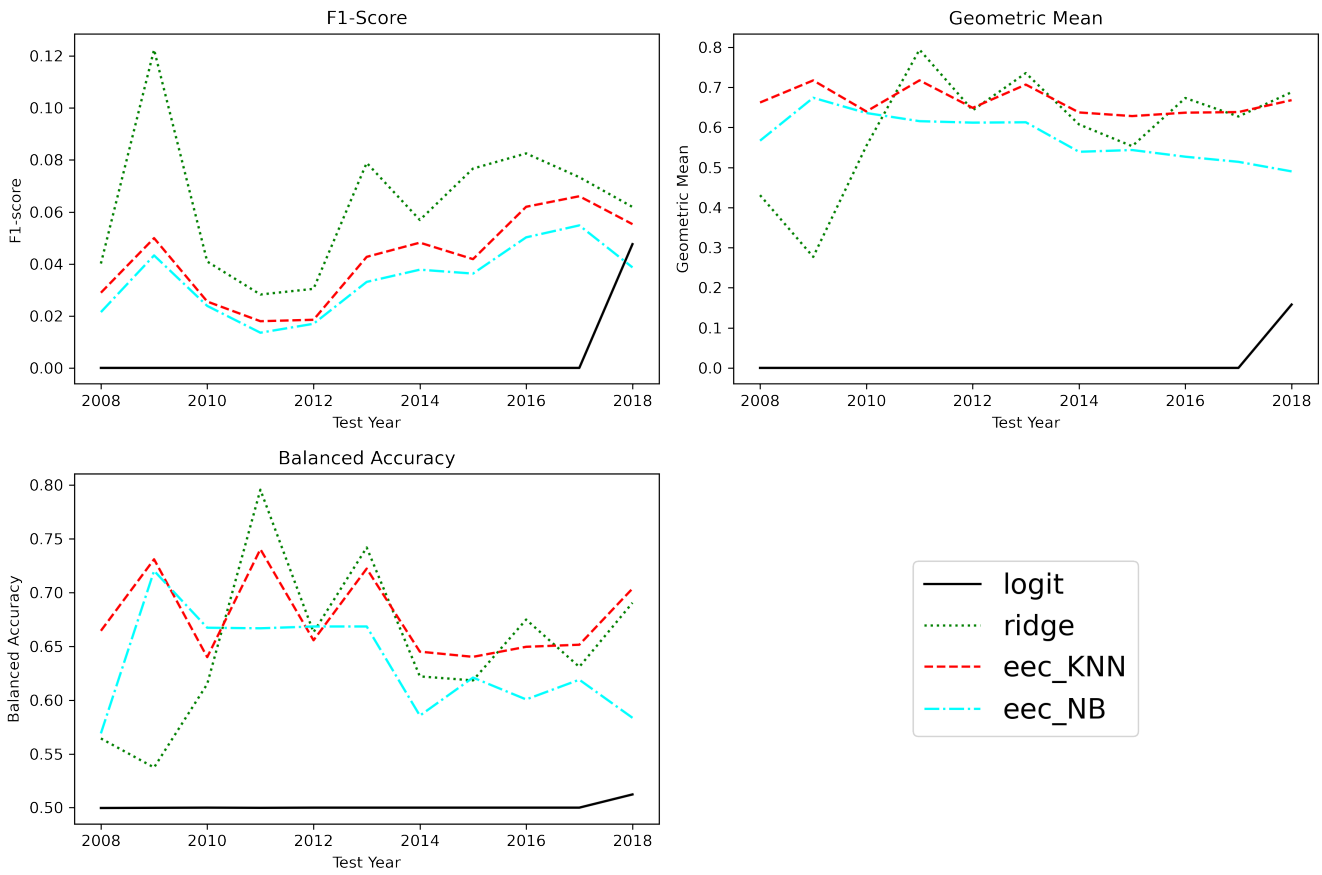


Fig. 7. Performance Metrics. The figure depicts the time series of three performance metrics for logit, ridge regression, EasyEnsemble based on KNN (EEC-KNN), and EasyEnsemble based on Naive Bayes (EEC-NB). The three metrics are: F1 score, G-mean, and Balanced Accuracy. F1-score is the harmonic mean between precision and recall. Geometric mean is the geometric mean between sensitivity and specificity. Balanced accuracy is the arthmetic mean between sensitivity and specificity. The black solid line indicates logit, the gray dotted line ridge, red dotted line EEC-KNN, and light blue dashed line EEC-NB. The metrics are all measured against test samples, and thus the sample runs from 2008 to 2018.

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Table 1: Summary Statistics
The table presents the summary statistics for key characteristics of both cyberattack victims and non-victims. There are in total 37,481 firm-year pairs, with 552 cyberattack victim cases, during the period 2006–2018. We show the mean, standard deviation, 25th percentile and 75th percentile for each group, and present them side by side for comparison. Victims are denoted as $Victim = 1$, while zero indicates non-victims. The variables are defined in the Appendix. Risk Factor Length is the number of tokens in a firm's 10-K Item 1A. NIST Term Counts is the number of times a firm mentions cybersecurity terms per NIST dictionary in its Item 1A in 10-K.

Variable	Victim	Mean	STD	P25	P75
Intangibility	0	0.79	0.23	0.70	0.97
	1	0.82	0.21	0.74	0.97
CAPX/AT	0	0.04	0.06	0.01	0.05
	1	0.03	0.04	0.01	0.05
CF/AT	0	0.02	0.22	0.01	0.11
	1	0.07	0.09	0.02	0.12
Firm Size	0	6.68	2.03	5.26	8.02
	1	9.22	2.30	7.56	10.99
Net Worth	0	0.45	0.25	0.25	0.66
	1	0.37	0.21	0.18	0.51
R&D	0	0.05	0.12	0.00	0.04
	1	0.02	0.06	0.00	0.01
ROA	0	-0.02	0.23	-0.01	0.07
	1	0.04	0.09	0.01	0.07
Tobin's Q	0	1.86	1.55	1.05	2.06
	1	1.86	1.19	1.09	2.10
Sales Growth	0	1.56	22.94	0.97	1.18
	1	1.10	0.23	1.00	1.13
Financial Constraint	0	-0.27	0.81	-0.33	-0.18
	1	-0.38	0.12	-0.47	-0.29
Annual Exret	0	0.01	0.52	-0.25	0.18
	1	0.03	0.31	-0.16	0.19
Log(age)	0	2.11	0.62	1.79	2.56
	1	2.26	0.60	1.95	2.71
Risk Factor Length	0	7,191	5,288	3,668	9,299
	1	8,710	6,422	4,304	12,019
NIST Counts	0	95	71	48	122
	1	135	96	66	178

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Table 2: Performance Metrics
This table reports the time-series mean out-of-sample performance metrics across the recursive windows for logit and ridge. F1-score is the harmonic mean between precision and recall. Geometric mean is the geometric mean between sensitivity and specificity. Balanced accuracy is the arthmetic mean between sensitivity and specificity. Each recursive window contains $n + 1$ years of data, with $n \in [1, 10]$ as the training data, and $+1$ as the subsequent year as the test data. There are in total 11 windows, starting at the year 2007 as the training data and 2008 as the test data. The final window consists of the year 2007 - 2017 as the training data and 2018 as the test data. The first two rows are the means of each corresponding metrics for 11 windows. The third row represents the mean difference between ridge and logit, and the fourth row computes the standard error of the difference in means.

Metrics	F1-score: Harmonic Mean	G-mean: Geometric Mean	Balanced Accuracy
Logit	0.004	0.014	0.501
Ridge	0.063	0.599	0.650
Ridge-Logit	0.059***	0.584***	0.149***
	(0.010)	(0.043)	(0.022)
Note:			*p<0.1; **p<0.05; ***p<0.01

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Table 3: Fama-MacBeth Cross-sectional Regressions

This table reports Fama-MacBeth cross-sectional regressions of returns on firm characteristics and anomaly variables. For each month, we run cross-sectional regression of stock returns on firm characteristics, and then we average the coefficients across time series. The sample runs from July 2008 to June 2019. Column (1) include natural log of market cap, natural log of book-to-market ratio, gross profitability, asset growth, and momentum; Column (2) adds short-term reversal, idiosyncratic risk (Ang et al., 2006), and illiquidity (Amihud, 2002); Column (3) adds organizational capital (Eisfeldt and Papanikolaou, 2013), past market beta, tail beta (Kelly and Jiang, 2014), coskewness (Harvey and Siddique, 2000), and net operating assets (Hirshleifer et al., 2004). Standard errors are Newey-West standard errors with 6 lags and are reported in parentheses. Variable definitions can be found in Appendix A. Returns are measured as percentages, and all independent variables are cross-sectionally winsorized at [1%,99%] level, and standardized to have zero mean and standard deviation of unity.

	Dependent Variable: Returns		
	(1)	(2)	(3)
Cyber Risk	0.197***	0.166**	0.208***
	(0.067)	(0.066)	(0.069)
Log(MktCap)	-0.127	-0.162	-0.235*
	(0.158)	(0.126)	(0.120)
Log(B/M)	0.163	0.087	0.071
	(0.148)	(0.147)	(0.141)
GP	0.098	0.076	0.376**
	(0.108)	(0.101)	(0.153)
ATG	-0.247***	-0.244***	-0.104
	(0.055)	(0.057)	(0.076)
MOM	-0.089	-0.116	-0.205
	(0.181)	(0.183)	(0.177)
ST-Rev		-0.341***	-0.379***
		(0.107)	(0.125)
IdioRisk		-0.222	-0.128
		(0.155)	(0.191)
Illiquidity		0.124	0.237*
		(0.103)	(0.135)
OrgCap			-0.240***
			(0.089)
Beta			-0.114
			(0.163)
TailBeta			0.090
			(0.070)
Coskew			0.033
			(0.053)
NOA			-0.184*
			(0.107)
Observations	289,696	289,696	138,950
R-squared	0.031	0.045	0.075
Number of groups	132	132	132

Note: *p<0.1; **p<0.05; ***p<0.01

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Table 4: Portfolio Analyses

This table provides various portfolio analyses. It presents portfolio high-minus-low alpha results from using terciles, quartiles, and quintiles, both value-weighted and equal-weighted, benchmarking to various asset pricing models, including CAPM, Fama-French three-factor model (FF3), FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6). The left half panel reports value-weighted results, while the right half panel reports equal-weighted results. Standard errors are Newey-West standard errors with 6 lags, and are included in parentheses.

	Value-Weighted			Equal-Weighted		
model	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile
CAPM	0.574***	0.765***	0.843***	0.508***	0.645***	0.713***
	(0.176)	(0.207)	(0.227)	(0.154)	(0.171)	(0.180)
FF3	0.458***	0.658***	0.739***	0.437***	0.581***	0.657***
	(0.122)	(0.190)	(0.219)	(0.131)	(0.154)	(0.157)
FF4	0.461***	0.660***	0.742***	0.433***	0.577***	0.653***
	(0.127)	(0.193)	(0.226)	(0.121)	(0.141)	(0.150)
FF5	0.431***	0.536***	0.577***	0.425***	0.541***	0.616***
	(0.116)	(0.157)	(0.179)	(0.138)	(0.160)	(0.165)
FF6	0.438***	0.540***	0.585***	0.419***	0.533***	0.611***
	(0.124)	(0.161)	(0.190)	(0.131)	(0.151)	(0.163)

Note: *p<0.1; **p<0.05; ***p<0.01

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Table 5: Cybersecurity ETFs: Top 10 Holdings
This table presents the top ten holdings of two cybersecurity ETFs respectively: First Trust Nasdaq Cybersecurity ETF (CIBR) and ETFMG Prime Cyber Security ETF (HACK). The holdings are as of February, 2021.

CIBR		HACK	
Holding	Weight	Holding	Weight
CrowdStrike Holdings, Inc. (Class A)	7.49%	Blackberry Ltd.	3.57%
Zscaler, Inc.	7.08%	Cisco Sys Inc.	3.13%
Cisco Systems, Inc.	5.60%	Palo Alto Networks Inc.	2.81%
Accenture Plc	5.25%	Cyberark Software Ltd.	2.80%
Splunk Inc.	4.41%	Ping Identity Hldg Corp	2.77%
FireEye, Inc.	3.41%	Fireeye Inc.	2.76%
Palo Alto Networks, Inc.	3.38%	Sumo Logic Inc.	2.71%
Proofpoint, Inc.	3.28%	Commvault Systems Inc.	2.69%
SailPoint Technologies Holdings, Inc.	3.27%	Cloudflare Inc.	2.64%
Fortinet, Inc.	3.22%	Qualys Inc.	2.64%

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Table 6: Sub-sample Analysis and Industry Effects

This table reports sub-sample analysis and industry analysis via Fama-MacBeth cross-sectional regressions. In the first two columns, we split our sample into two sub-periods, from July 2008 to December 2013, and from January 2014 to June 2019. We regress monthly stock returns on the common stock characteristics including natural log of market cap, natural log of book-to-market ratio, gross profitability, asset growth, and momentum. And report the time-series means of the coefficients for the two sub-periods in Column (1) and (2) separately. In the third column, we conduct industry analysis to see whether our results are driven by industry returns. In each month in our sample, we subtract Fama-French 48 industry (Fama and French, 1997) portfolio returns from stock returns to arrive at the industry-asjusted excess returns. Then we conduct Fama-MacBeth regressions using the adjusted returns on the stock characteristics, and report the time-series means in Column (3). Standard errors are Newey-West standard errors with 6 lags and are reported in parentheses. Variable definitions can be found in Appendix. Returns are measured as percentages, and all independent variables are cross-sectionally winsorized at [1%,99%] level, and standardized to have zero mean and standard deviation of unity.

	(1)	(2)	(3)
	2008/07-	2014/01-	FF48-Industry
VARIABLES	2013/12	2019/06	Adjusted Returns
Cyber Risk	0.180*	0.214**	0.145***
	(0.095)	(0.094)	(0.054)
Log(MktCap)	-0.298	0.044	-0.058
	(0.254)	(0.176)	(0.147)
Log(B/M)	0.363*	-0.037	0.205
	(0.212)	(0.191)	(0.129)
GP	0.053	0.144	0.047
	(0.102)	(0.191)	(0.104)
ATG	-0.334***	-0.160**	-0.205***
	(0.070)	(0.076)	(0.043)
MOM	-0.223	0.046	-0.117
	(0.326)	(0.147)	(0.167)
Constant	1.669	0.563	0.113
	(1.157)	(0.577)	(0.288)
Observations	153,509	136,187	276,025
R-squared	0.026	0.035	0.026
Number of groups	66	66	132
Note:			*p<0.1; **p<0.05; ***p<0.01

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Table 7: Portfolio Analyses on Ensemble-based Cyber Risk
This table provides portfolio analyses by using cyber risk estimated by EasyEnsemble methods using K-nearest neighbors and Naive Bayes. We show portfolio high-minus-low alpha results from using terciles, quartiles, and quintiles, from both EEC-KNN and EEC-NB. These are benchmarked to various asset pricing models, including CAPM, Fama-French three-factor model (FF3), FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6). The left half panel reports results from using EEC-KNN generated cyber risk, while the right half panel reports results from using EEC-NB generated cyber risk. Standard errors are Newey-West standard errors with 6 lags, and are included in parentheses.

	EEC-KNN			EEC-NB		
model	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile
CAPM	0.467***	0.556***	0.489***	0.509***	0.711***	0.984***
	(0.164)	(0.198)	(0.204)	(0.262)	(0.267)	(0.273)
FF3	0.366***	0.446***	0.381***	0.606***	0.812***	1.058***
	(0.091)	(0.088)	(0.099)	(0.228)	(0.209)	(0.228)
FF4	0.369***	0.449***	0.385***	0.616***	0.824***	1.072***
	(0.094)	(0.091)	(0.102)	(0.239)	(0.219)	(0.252)
FF5	0.270***	0.335***	0.239**	0.355**	0.537***	0.744***
	(0.083)	(0.092)	(0.099)	(0.179)	(0.157)	(0.162)
FF6	0.276***	0.342***	0.246**	0.372**	0.559***	0.770***
	(0.085)	(0.094)	(0.101)	(0.182)	(0.160)	(0.176)

Note: *p<0.1; **p<0.05; ***p<0.01

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Table 8: Placebo "Pseudo-Cyber-Risk" Portfolio High-Minus-Low Alphas
This table reports the placebo portfolio high-minus-low alphas. First, for each hacked firm in each year, we randomly sample a firm from the same Fama-French 48 industry, and label it as the "pseudo" attacked firm; we do this for all the attacked firms in our sample, and thus we have 394 "pseudo" attacked firms. We follow the same prediction procedure as in our main results (logistic ridge regression), but on the sample of these "pseudo" attacked firms, and generate a "pseudo" cyber risk measure for each firm. We finally test whether by using this "pseudo" measure we can find significant high-minus-low portfolio alphas. We perform these tests on terciles, quartiles, and quintile sorts based on this pseudo measure, and then regress the resulting monthly return series on various empirical asset pricing factors. Standard errors are adjusted with Newey-West procedure with 6 lags, and are included in parentheses.

model	Tercile	Quartile	Quintile
CAPM	-0.130	-0.198	-0.210
	(0.216)	(0.286)	(0.325)
FF3	-0.162	-0.227	-0.273
	(0.242)	(0.316)	(0.350)
FF4	-0.161	-0.225	-0.272
	(0.243)	(0.317)	(0.351)
FF5	-0.212	-0.293	-0.344
	(0.229)	(0.297)	(0.334)
FF6	-0.214	-0.293	-0.347
	(0.230)	(0.299)	(0.335)
Note:			*p<0.1; **p<0.05; ***p<0.01

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Appendix A. Selected Variable Definitions

ACI	= CAPX ratio increase over the previous three periods mean. CAPX ratio is $CAPX/SALE$.
AG	= asset growth over the previous year
$BookValueofEquity$	= $SEQ + TXDITC - Perferred$, preferred is $PSTKRV$, or $PSTKL$, or $PSTK$, whichever is first available.
$CAPX/AT$	= $CAPX/AT$
CF/AT	= $(IB + DP)/AT$
$Cyber\ Risk$	= predicted one-year ahead probability of suffering cyberattack events
$FinancialConstraint$	= $-0.091 \times CF/AT - 0.062 \times DIVPOS + 0.021 \times TLTD - 0.044 \times Firm\ Size + 0.102 \times SIC - 3digit - industry - growth - 0.035 \times Sale\ Growth$ (Whited and Wu (2006))
$Firm\ Size$	= natural log of firm's total assets
$F1 - score$	= $(precision \times recall)/(precision + recall)$
$G - mean$	= $\sqrt{sensitivity \times specificity}$
GP	= gross profitability, equals $(REVT - COGS)/AT$
$Illiquidity$	= monthly mean of daily absolute return over price times volume of that day, see Amihud (2002).
$Intangibility$	= $1 - PPENT/AT$
$Net\ Operating\ Assets$	= $AT - CHE - (AT - DLC - DLTT - MIB - PSTK - CEQ)$
$Net\ Worth$	= SEQ/AT
NOA	= $net_operating_assets/lag_AT$
NSI	= natural log of changes in adjusted shares
$OSCR$	= $-1.32 - 0.407 \times ASIZE + 6.03 \times TLTA - 1.43 \times WCTA + 0.0757 \times CLCA - 1.72 \times OENEG - 2.37 \times NITA - 1.83 \times FUTL + 0.285 \times INTWO - 0.521 \times CHIN$, O-score, see Ohlson (1980).
$R\&D$	= max of XRD/AT or zero
ROA	= NI/AT
$Tobin's\ Q$	= $(AT - CEQ + PRCC_F \times CSHO)/AT$

Appendix B. In-sample Logit

This section provides detailed results for the in-sample logit produced in Section 3. The results are shown in Table A.1.

[Table A.1 about here.]

As expected and consistent with findings in Kamiya et al. (2021), larger firms (as measured by the log of firm's total assets), firms with more intangible assets, higher cash flows, more growth po-

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tential (measured by Tobin's q), and firms that face more financial constraints (measured according to Whited and Wu (2006)), are more likely victims of cyber attacks next year. On the other hand, firms with low return on assets are more likely victims.

Appendix C. Model Results with Different Cyberattack Screening

In our main results, we select "important" cyberattack incidents as our response variable to predict future likelihood of cyberattacks. The screening is based on whether the [-3,3] window cumulative abnormal returns of the underlying stocks who are cyberattack victims cross the threshold of 1% in absolute value. Here we show that the results are robust to setting no threshold, or setting 0.5% as the screening threshold. Table A.2 and Table A.3 report results from using no screening and using 0.5% as screening threshold respectively.

[Table A.2 about here.]

[Table A.3 about here.]

From the two tables, we can see that, with a few exceptions, the high-minus-low cyber risk portfolio alpha estimates of all models cross the T-statistic threshold of 3, indicating that our results are robust to setting different screening standards for the cyberattack events.

Appendix D. Alternative Dictionary Results

In this section we discuss the results from using the dictionary described in Jamilov et al. (2020). Their dictionary of cyber risk is reproduced here:"cyber", "cybersecurity", "network security", "cyberattack", "cybercrime", "cyber threat", "cyber incident", "cyber event", "data loss", "data integrity", "data security", "information theft", "data breach", "data theft", "data leak", "data compromise", "data fraud", "worm", "spyware", "phishing", "trojan", "malware", "ddos attack", "ransomware", "hacker", "hack", "hacked", "card fraud", "card breach", "system outage", "email compromise". We find that there are similarities and distinctions between our dictionary (NIST) and this dictionary. NIST dictionary has 578 terms in total, while the dictionary in Jamilov et al. (2020) has 31 terms. There are 5 overlapping terms between the two: "data integrity", "malware", "phishing", "spyware", "worm". Overall, the two dictionaries are different in their coverage and depth.

Next, we count the number of times these words and phrases described in Jamilov et al. (2020) are mentioned in each firm's Item 1A "Risk Factors" section of their 10-K filing, and substitute our NIST variable with this new variable. Then we follow the same procedure of expanding and

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recursive predictions via logit and machine learning. We plot the resulting performance metrics in Figure A.1.

[Fig. A.1 about here.]

Next we present the results from portfolio sorts. We sort stocks at the end of each year by estimated probability of cyberattack using the aforementioned results, and regress the value-weighted portfolio excess returns on various factors per asset pricing models. We report the resulting alpha estimates and associated Newey-West standard errors estimated with 6 lags in Table A.4.

[Table A.4 about here.]

As shown in the table, all alpha estimates are economically and statistically significant and positive. This supports the notion that our methodology is flexible and robust to using alternative cybersecurity dictionaries, and that cyber risk is priced in the cross section of common stocks.

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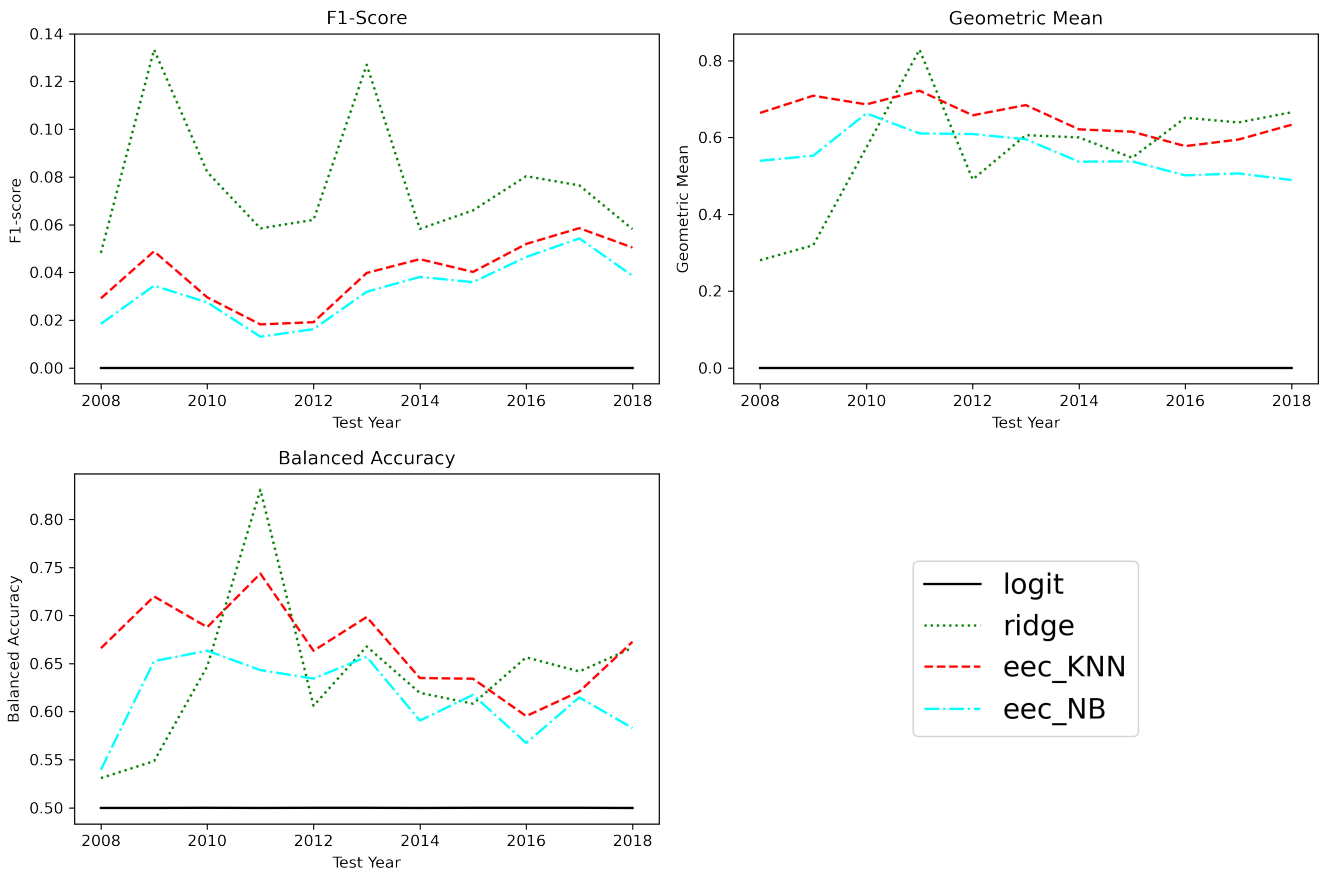


Fig. A.1. Performance Metrics for Models Using Alternative Dictionary. The figure depicts the time series of three performance metrics for logit, ridge regression, EasyEnsemble based on KNN (ECC-KNN), and EasyEnsemble based on Naive Bayes (EEC-NB). These models use Jamilov et al. (2020) dictionary of cybersecurity instead of the NIST glossary used in the main results. The three metrics are: F1 score, G-mean, and Balanced Accuracy. F1-score is the harmonic mean between precision and recall. Geometric mean is the geometric mean between sensitivity and specificity. Balanced accuracy is the arthmetic mean between sensitivity and specificity. The black solid line indicates logit, the gray dotted line ridge, red dotted line EEC-KNN, and light blue dashed line EEC-NB. The metrics are all measured against test samples, and thus the sample runs from 2008 to 2018.

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Table A.1: In-Sample Logit

This table reports in-sample logit regressions incorporating the set of disclosure variables for cyber risk. The dependent variable is a dummy that equals one if the firm experienced cyberattack in the year, or zero otherwise. Explanatory variables are winsorized at [1%, 99%] level and standardized to be mean zero and standard deviation of unity to compare the relative importance among the explanatory variables. Standard errors are clustered at both firm level and year level. Sample runs from 2006 to 2018. Risk Factor Length measures the number of tokens in each risk factor section. The "count" variables are term counts per the specific cybersecurity dictionary. $NIST$ refers to NIST dictionary, $NICCS$ NICCS dictionary, and $Cyber$ CyberPolicy dictionary.

VARIABLES	Dependent Variable: Cyberattack		
Intangibility	0.314***	0.376***	0.353***
	(0.094)	(0.094)	(0.094)
CAPX/AT	0.088	0.084	0.080
	(0.095)	(0.095)	(0.095)
CF/AT	1.130***	1.121***	1.102***
	(0.246)	(0.250)	(0.255)
Firm Size	1.549***	1.571***	1.569***
	(0.095)	(0.100)	(0.097)
Net Worth	0.194***	0.167**	0.170**
	(0.068)	(0.066)	(0.067)
R&D	-0.130	-0.198	-0.202
	(0.151)	(0.165)	(0.168)
ROA	-1.042***	-1.042***	-1.022***
	(0.260)	(0.262)	(0.268)
Tobin's Q	0.226***	0.248***	0.245***
	(0.048)	(0.048)	(0.049)
Sales Growth	-0.039	-0.056	-0.056
	(0.071)	(0.070)	(0.069)
Financial Constraint	0.333***	0.298***	0.310***
	(0.088)	(0.090)	(0.091)
Annual Exret	0.036	0.041	0.040
	(0.062)	(0.063)	(0.063)
Log(age)	-0.006	0.020	0.025
	(0.039)	(0.043)	(0.039)
Risk Factor Length	-0.393***	0.041	-0.006
	(0.097)	(0.124)	(0.089)
NIST Counts	0.590***		
	(0.091)		
NICCS Counts		0.171	
		(0.111)	
Cyber Counts			0.223***
			(0.075)
Observations	37,481	37,481	37,481
Pseudo R2	0.163	0.157	0.158

Note: *p<0.1; **p<0.05; ***p<0.01

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Table A.2: Portfolio Alphas by Forecasting Models: No Event Screening

This table presents the alpha estimates and associated standard errors from regressing high-minus-low cyber risk portfolio returns on various risk factors per the pricing model used, using all cyberattack incidents without screening. The results reflect different cyber risk estimates from using different forecasting models: ridge, EEC-KNN, and EEC-NB. The pricing models include: CAPM, Fama-French three-factor model (FF3), FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6). Standard errors are Newey-West standard errors with 6 lags. Sample runs from January 2008 to December 2019.

	Ridge			EEC-KNN			KNN-NB		
model	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile
CAPM	0.542***	0.661***	0.695***	0.421**	0.456**	0.523**	0.286	0.493**	0.633**
	(0.185)	(0.222)	(0.248)	(0.209)	(0.212)	(0.238)	(0.215)	(0.245)	(0.273)
FF3	0.462***	0.585***	0.635***	0.293***	0.355***	0.447***	0.414**	0.653***	0.785***
	(0.128)	(0.182)	(0.207)	(0.108)	(0.096)	(0.119)	(0.194)	(0.212)	(0.221)
FF4	0.464***	0.584***	0.633***	0.283***	0.351***	0.445***	0.406**	0.642***	0.777***
	(0.125)	(0.183)	(0.208)	(0.109)	(0.098)	(0.122)	(0.195)	(0.214)	(0.223)
FF5	0.367***	0.400**	0.398**	0.220**	0.233**	0.290**	0.224	0.402**	0.510***
	(0.113)	(0.151)	(0.167)	(0.106)	(0.105)	(0.131)	(0.159)	(0.174)	(0.183)
FF6	0.368***	0.400**	0.398**	0.215**	0.232**	0.289**	0.221	0.398**	0.507***
	(0.110)	(0.151)	(0.168)	(0.108)	(0.106)	(0.134)	(0.158)	(0.171)	(0.182)

Note: *p<0.1; **p<0.05; ***p<0.01

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Table A.3: Portfolio Alphas by Forecasting Models: Event Screening by 0.5% CAR

This table presents the alpha estimates and associated standard errors from regressing high-minus-low cyber risk portfolio returns on various risk factors per the pricing model used, using only cyberattack incidents whose [-3,3] cumulative abnormal returns cross the threshold of 0.5% in absolute value. The results reflect different cyber risk estimates from using different forecasting models: ridge, EEC-KNN, and EEC-NB. The pricing models include: CAPM, Fama-French three-factor model (FF3), FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6). Standard errors are Newey-West standard errors with 6 lags. Sample runs from January 2008 to December 2019.

	Ridge			EEC-KNN			KNN-NB		
model	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile
CAPM	0.551**	0.684***	0.753***	0.487***	0.548***	0.653***	0.568**	0.619**	0.797**
	(0.234)	(0.254)	(0.278)	(0.174)	(0.201)	(0.210)	(0.252)	(0.303)	(0.326)
FF3	0.546***	0.673***	0.744***	0.375***	0.436***	0.537***	0.665***	0.740***	0.913***
	(0.160)	(0.168)	(0.170)	(0.094)	(0.097)	(0.102)	(0.220)	(0.244)	(0.235)
FF4	0.548***	0.676***	0.746***	0.380***	0.440***	0.540***	0.676***	0.752***	0.925***
	(0.159)	(0.168)	(0.170)	(0.095)	(0.101)	(0.105)	(0.229)	(0.243)	(0.231)
FF5	0.486***	0.561***	0.596***	0.296***	0.313***	0.391***	0.423**	0.440**	0.597***
	(0.142)	(0.139)	(0.136)	(0.084)	(0.090)	(0.100)	(0.173)	(0.190)	(0.184)
FF6	0.493***	0.568***	0.602***	0.304***	0.320***	0.398***	0.443**	0.462**	0.619***
	(0.136)	(0.135)	(0.132)	(0.085)	(0.094)	(0.103)	(0.171)	(0.177)	(0.164)

Note: *p<0.1; **p<0.05; ***p<0.01

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Table A.4: Portfolio Alphas by Forecasting Models: Alternative Dictionary

This table presents the alpha estimates and associated standard errors from regressing high-minus-low cyber risk portfolio returns on various risk factors per the pricing model used, by using cyber risk dictionary from Jamilov et al. (2020). The results reflect different cyber risk estimates from using different forecasting models: ridge, EEC-KNN, and EEC-NB. The pricing models include: CAPM, Fama-French three-factor model (FF3), FF3 augmented by a momentum factor (FF4), Fama-French five-factor model (FF5), and FF5 augmented with a momentum factor (FF6). Standard errors are Newey-West standard errors with 6 lags. Sample runs from January 2008 to December 2019.

	Ridge			EEC-KNN			KNN-NB		
model	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile	Tercile	Quartile	Quintile
CAPM	0.784***	0.932***	1.006***	0.479***	0.424***	0.424***	0.503***	0.743***	0.920***
	(0.245)	(0.285)	(0.308)	(0.172)	(0.199)	(0.184)	(0.252)	(0.289)	(0.313)
FF3	0.829***	1.000***	1.097***	0.393***	0.349***	0.345***	0.606***	0.883***	1.067***
	(0.243)	(0.269)	(0.292)	(0.078)	(0.092)	(0.108)	(0.205)	(0.231)	(0.265)
FF4	0.838***	1.009***	1.107***	0.395***	0.351***	0.348***	0.616***	0.894***	1.079***
	(0.251)	(0.278)	(0.303)	(0.079)	(0.094)	(0.112)	(0.214)	(0.241)	(0.278)
FF5	0.688***	0.834***	0.907***	0.305***	0.241**	0.221*	0.352**	0.570***	0.766***
	(0.208)	(0.223)	(0.240)	(0.082)	(0.098)	(0.115)	(0.163)	(0.162)	(0.200)
FF6	0.705***	0.853***	0.927***	0.308***	0.244**	0.224*	0.368**	0.591***	0.789***
	(0.209)	(0.227)	(0.248)	(0.081)	(0.099)	(0.118)	(0.165)	(0.165)	(0.207)

Note: *p<0.1; **p<0.05; ***p<0.01

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